

Question 1 (15 Marks)

Marks

(a) Integrate with respect to x :

(i) $x^2 - \sqrt{x}$

2

(ii) $\sin 8x$

2

(iii) $\frac{6}{1-3x}$

2

(b) Evaluate:

(i) $\int_0^3 e^{2x+1} dx$

2

(ii) $\int_0^\pi \cos\left(\frac{1}{4}x\right) dx$

2

(iii) $\int_1^4 \frac{2x^2 - 3}{x} dx$

2

(c) (i) Evaluate $\sum_{r=1}^5 r^2$

1

(ii) Find the sum of the first 200 terms of the arithmetic series: $3 + 7 + 11 + 15 + \dots$

2

Question 2 (15 Marks)

Marks

(a) Find:

(i) $\int (3x - 2)^8 dx$

2

(ii) $\int \frac{e^{3x}}{5 + e^{3x}} dx$

2

(b) (i) Sketch the curve $y = 6x - x^2$. Clearly show its intercepts with the coordinate axes.

2

(ii) Find the area bounded by the curve $y = 6x - x^2$ and the x -axis.

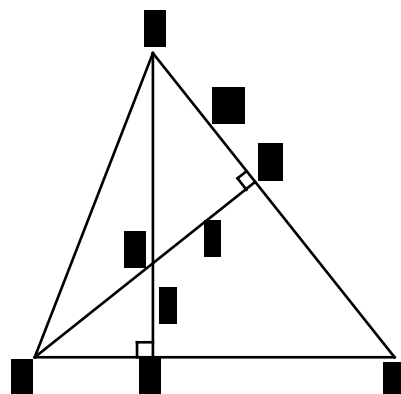
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(c) In $\triangle ABC$, $AP \perp BC$ and $BQ \perp AC$.(i) Prove that $\triangle AQR \parallel \triangle BPR$.

3

(ii) If $AQ = 12$, $RQ = 9$ and $RP = 6$, find the area of $\triangle ABR$.

3



Question 3 (15 Marks)

Marks

- (a) The area bounded by the curve $y = \sqrt{4-x}$ and the coordinate axes is rotated one revolution about the y-axis. Find the volume of the solid formed. 3
- (b) (i) Show that the curves $y = \cos x$ and $y = \sin 2x$ meet at a point where $x = \frac{\pi}{6}$. 2
- (ii) On the same set of axes draw a neat half page diagram of the curves $y = \cos x$ and $y = \sin 2x$ for $0 \leq x \leq \frac{\pi}{2}$ clearly showing their points of intersection. 2
- (iii) Find the area bounded by the above curves for $0 \leq x \leq \frac{\pi}{2}$. 3
- (c) Mr. Green sets up a fund to pay for a future holiday by paying \$500 into the fund at the start of each month. The fund pays 0.5% interest at the end of each month on the balance of money in the fund.
- (i) Find the value of the fund at the end of the first year. 2
- (ii) How long must Mr. Green pay into the fund if he needs \$20000 for his holiday. Give your answer to the nearest month. 3

Question 4 (15 Marks)

Marks

- (a) An artist decides to make a design showing a sequence of concentric rope circles. The inner most circle has a radius of 40 cm and each extra circle has a radius that is 10 cm larger than the radius of the previous circle.
- (i) Find the amount of rope used to make the tenth circle. Give your answer to the nearest metre. 2
- (ii) If the artist has 2 km of rope, how many complete circles will be in the design? 3
- (b) (i) Given that $f(x) = \ln(\cos 2x)$, find $f'(x)$. 1
- (ii) Prove that the equation of the tangent to $y = \tan 2x$ at the point $P(\frac{\pi}{8}, 1)$ has the equation $y = 4x + 1 - \frac{\pi}{2}$. 3
- (iii) Find the coordinates of Q , the point where the tangent at P crosses the x-axis. 1
- (iv) On a neat half page diagram, sketch the curve $y = \tan 2x$ for $0 \leq x \leq \frac{\pi}{4}$ and the tangent at P . 2
- (v) Find the area bounded by the curve $y = \tan 2x$, the tangent at P and the x-axis. Express your answer in the form $a + b \ln(2)$, where a and b are rational numbers. 3

This is the end of the Examination Paper

SOLUTIONS

Question 1 (15 Marks)

Marks

(a) Integrate with respect to x :

(i) $\frac{1}{3}x^3 - \frac{2}{3}x\sqrt{x} + c$

each part = 1 mark

2

- 1 for no “c” only once in part (a)

(ii) $-\frac{1}{8}\cos 8x + c$

 $\cos 8x = 1$ mark

2

-1/8 = 1 mark

(iii) $-2\ln(1-3x) + c$ or $-2\ln|1-3x| + c$

 $\ln(1-3x) = 1$ mark

2

-2 = 1 mark

(b) Evaluate:

(i) $\int_0^3 e^{2x+1} dx = \left[\frac{1}{2} e^{2x+1} \right]_0^3$
 $= \frac{1}{2}(e^7 - e^1)$

primitive = 1 mark

2

evaluation = 1 mark

(ii) $\int_0^\pi \cos\left(\frac{1}{4}x\right) dx = \left[4\sin\left(\frac{1}{4}x\right) \right]_0^\pi$
 $= 4\left(\sin\left(\frac{\pi}{4}\right) - \sin 0\right)$
 $= 4 \times \frac{1}{\sqrt{2}} - 0$
 $= 2\sqrt{2}$

primitive = 1 mark

2

evaluation = 1 mark

(iii) $\int_1^4 \frac{2x^2 - 3}{x} dx = \int_1^4 \left(2x - \frac{3}{x} \right) dx$
 $= \left[x^2 - 3\ln x \right]_1^4$
 $= (16 - 3\ln 4) - (1 - 3\ln 1)$
 $= 15 - 3\ln 4$

primitive = 1 mark

2

evaluation = 1 mark

(c) (i) $\sum_{r=1}^5 r^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2$
 $= 55$

accept correct answer with no working

1

(ii) $n = 200, a = 3, d = 4$

2

$$S_{200} = \frac{200}{2} \{2 \times 3 + 199 \times 4\}$$
$$= 80200$$

sub. into correct formula = 1 mark

correct sum = 1 mark

Question 2 (15 Marks)

Marks

(a) Find:

$$(i) \int (3x-2)^8 dx = \frac{1}{27}(3x-2)^9 + c$$

 $(3x-2)^9 = 1$ mark
no penalty for missing "c"

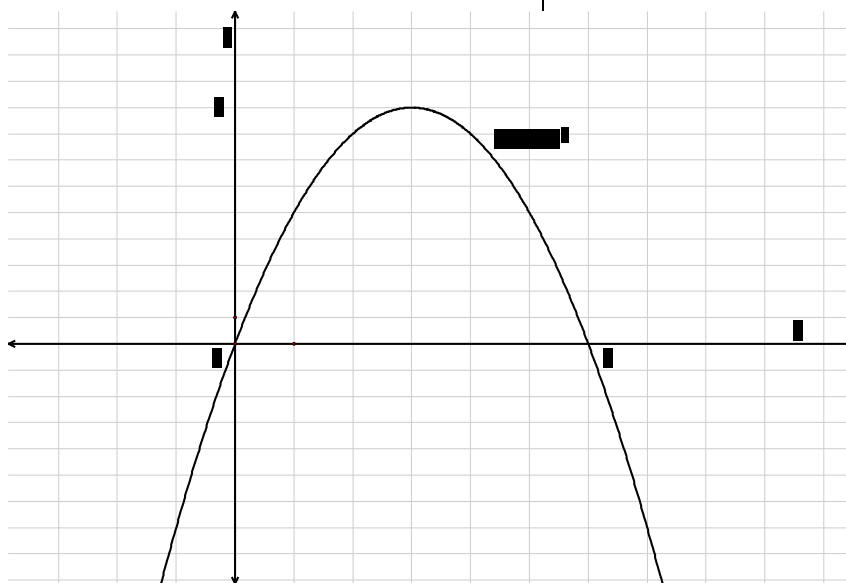
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$$(ii) \int \frac{e^{3x}}{5+e^{3x}} dx = \frac{1}{3} \ln(5+e^{3x}) + c$$

 $\ln(5+e^{3x}) = 1$ mark
no penalty for missing "c"

2

(b) (i)



2

$$(ii) \begin{aligned} A &= \int_0^6 (6x - x^2) dx \\ &= \left[3x^2 - \frac{1}{3}x^3 \right]_0^6 \\ &= \left(3 \times 6^2 - \frac{1}{3} \times 6^3 \right) - (0 - 0) \\ &= 36 \\ \therefore \text{Area} &= 36 u^2 \end{aligned}$$

both intercepts = 1 mark
shape = 1 mark

correct integral = 1 mark

3

integration = 1 mark

evaluation = 1 mark

max. 2/3 if no units

(c) In $\triangle ABC$, $AP \perp BC$ and $BQ \perp AC$.

(i)

In $\triangle AQR \parallel \triangle BPR$

$$\hat{AQR} = \hat{BPR} \quad (\text{both } 90^\circ)$$

$$\hat{ARQ} = \hat{BRP} \quad \left(\begin{array}{l} \text{vertically opposite} \\ \text{angles are equal} \end{array} \right)$$

$$\therefore \triangle AQR \parallel \triangle BPR \quad (\text{equiangular})$$

(ii)

$$\text{Area } \triangle ABR = \frac{BR \times AQ}{2}$$

$$\frac{BP}{12} = \frac{6}{9} \quad \left(\begin{array}{l} \text{ratio of corresponding} \\ \text{sides in similar triangles} \\ \text{are equal} \end{array} \right)$$

$$BP = 8$$

$$BR^2 = 8^2 + 6^2 \quad (\text{Pythagoras' Theorem})$$

$$BR = 10$$

$$\begin{aligned} \text{Area} &= \frac{10 \times 12}{2} u^2 \\ &= 60 u^2 \end{aligned}$$

3

angle pairs (= 1 mark each) = 2 marks

test = 1 mark

3

calculating 2 needed lengths (= 1 mark each) = 2 marks

several pairs of sides and various triangle combinations can be used to find the required area

area = 1 mark

Question 3 (15 Marks)

Marks

(a) $V = \pi \int_0^2 x^2 dy$ where $y = \sqrt{4-x}$

$$y^2 = 4 - x$$

$$x = 4 - y^2$$

$$V = \pi \int_0^2 (4 - y^2)^2 dy$$

$$= \pi \int_0^2 (16 - 8y^2 + y^4) dy$$

$$= \pi \left[16y - \frac{8}{3}y^3 + \frac{1}{5}y^5 \right]_0^2$$

$$= \pi \left\{ \left(16 \times 2 - \frac{8}{3} \times 8 + \frac{1}{5} \times 32 \right) - 0 \right\}$$

$$= \frac{256\pi}{15}$$

$$\therefore \text{Volume} = \frac{256\pi}{15} u^3$$

integral = 1 mark

wrong integrand = max 1/3

integration = 1 mark

evaluation = 1 mark

no penalty for missing units

(b) (i) when $x = \frac{\pi}{6}$ $\cos x = \cos \frac{\pi}{6}$

$$= \frac{\sqrt{3}}{2}$$

$$\sin 2x = \sin \frac{2\pi}{6}$$

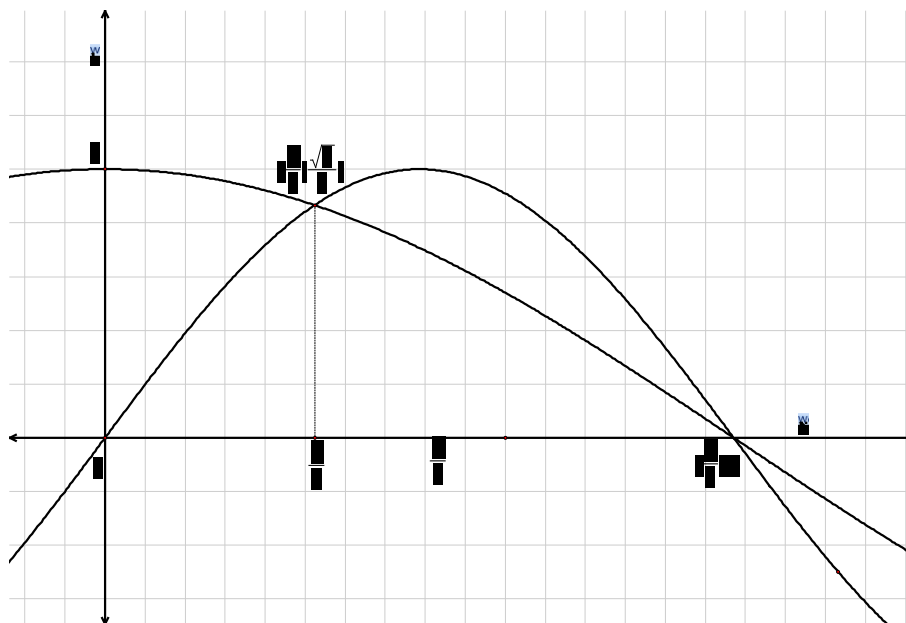
$$= \sin \frac{\pi}{3}$$

$$= \frac{\sqrt{3}}{2}$$

$$\therefore \text{both curves meet at } \left(\frac{\pi}{6}, \frac{\sqrt{3}}{2} \right)$$

show that each y value = $\frac{\sqrt{3}}{2}$
(= 1 mark each) = 2 marks

(ii)



1 mark for each curve

no penalty if domain extends beyond $\frac{\pi}{2}$

no penalty for missing intersection point

(iii) $A = \int_{\pi/6}^{\pi/2} (\sin 2x - \cos x) dx$ $= \left[-\frac{1}{2} \cos 2x - \sin x \right]_{\pi/6}^{\pi/2}$ $= \left(-\frac{1}{2} \cos \pi - \sin\left(\frac{\pi}{2}\right) \right) - \left(-\frac{1}{2} \cos\left(\frac{2\pi}{6}\right) - \sin\left(\frac{\pi}{6}\right) \right)$ $= \frac{1}{4}$ $\therefore \text{Area} = \frac{1}{4} u^2$	integral = 1 mark integration = 1 mark evaluation = 1 mark no penalty for missing units 3
(c) (i) Let value of fund at end of n^{th} month = $\\$A_n$ $A_1 = 500 \times 1.005$ $A_2 = (A_1 \times 1.005 + 500) \times 1.005$ $= 500(1.005^2 + 1.005)$ $\vdots$ $\vdots$ $A_{12} = 500(1.005^{12} + 1.005^{11} + \dots + 1.005)$ $= 500 \times 1.005 \frac{(1.005^{12} - 1)}{1.005 - 1}$ $= 6167.78$ Value = \$6167.78	2 correct expression = 1 mark evaluation of sum = 1 mark
(ii) $A_n = 500(1.005^n + 1.005^{n-1} + \dots + 1.005)$ $= 500 \times 1.005 \frac{(1.005^n - 1)}{0.005}$ $= 100500(1.005^n - 1)$ $\therefore 20000 = 100500(1.005^n - 1)$ $1.005^n = \frac{241}{201}$ $n \ln(1.005) = \ln\left(\frac{241}{201}\right)$ $n = \frac{\ln\left(\frac{241}{201}\right)}{\ln(1.005)}$ ≈ 36.39 no. months = 36	correct general expression = 1 mark 3 simplified equation = 1 mark solve equation = 1 mark (by trial and error or logs) accept 37 months

Question 4 (15 Marks)

Marks

(a) (i) circumferences: $\{80\pi, 100\pi, 120\pi, \dots\}$
 $a = 80\pi, d = 20\pi$
 $T_{10} = 80\pi + 9 \times 20\pi$
 $= 260\pi$

Length = 260π cm
 ≈ 816.81 cm
 $= 8m$ (to nearest metre)

(ii) $S_n = \frac{n}{2} \{2 \times 80\pi + (n-1) \times 20\pi\}$
 when $S_n = 200000$
 $\frac{n}{2} \{2 \times 80\pi + (n-1) \times 20\pi\} = 200000$
 $\pi n^2 + 7\pi n - 20000 = 0$
 $n = \frac{-7\pi \pm \sqrt{80049\pi}}{2\pi}$
 $n \approx -83.2$ or 76.3
 but $n > 0$
 $\therefore$ no. complete circles = 76

(b) (i) $f(x) = \ln(\cos 2x)$
 $f'(x) = \frac{-2 \sin 2x}{\cos 2x}$
 $= -2 \tan 2x$

(ii) $y = \tan 2x$
 $y' = 2 \sec^2 2x$
 when $x = \frac{\pi}{8}$ $y' = 2 \sec^2\left(\frac{\pi}{4}\right)$
 $= 2(\sqrt{2})^2$
 $= 4$
 tangent is: $y - 1 = 4\left(x - \frac{\pi}{8}\right)$
 $y - 1 = 4x - \frac{\pi}{2}$
 $y = 4x + 1 - \frac{\pi}{2}$

(iii) at Q , $y = 0$
 $\therefore 4x + 1 - \frac{\pi}{2} = 0$
 $x = \frac{\pi}{8} - \frac{1}{4}$
 Q is $\left(\frac{\pi}{8} - \frac{1}{4}, 0\right)$

2

indication of correct “a” and “d” = 1 mark

evaluate length = 1 mark
(using their “a” and “d”)

correct decimal approx. = 1 mark

3

sum formula with substitution = 1 mark

simplified quadratic equation = 1 mark

correct approximation = 1 mark

1

3

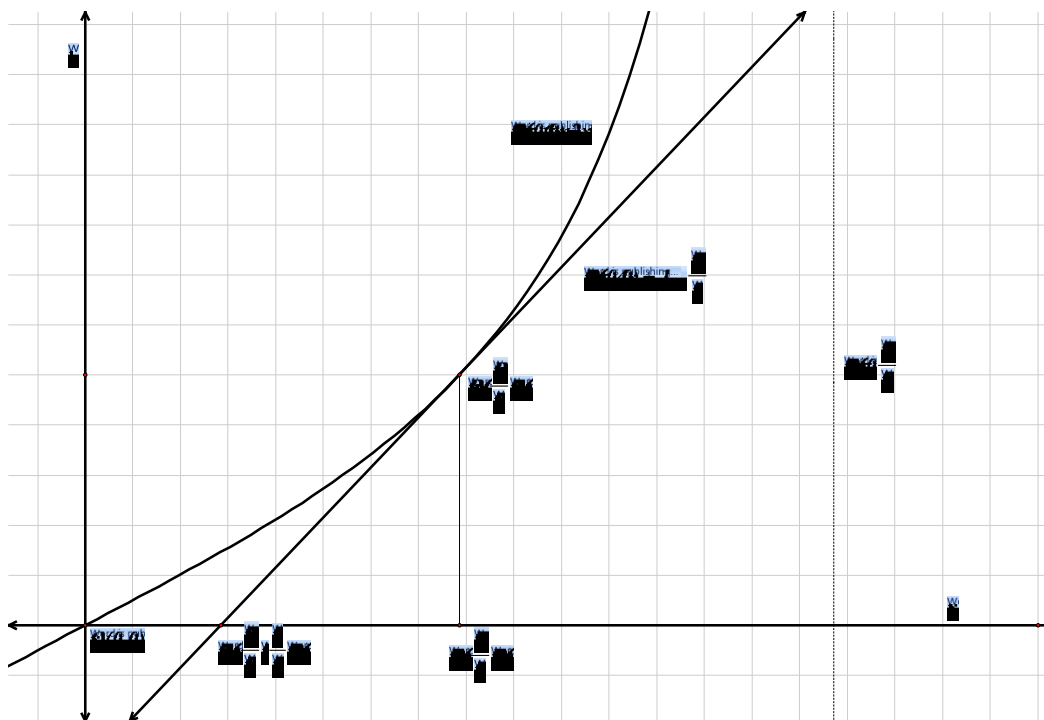
derivative = 1 mark

gradient = 1 mark

tangent equation = 1 mark

1

(iv)



2

tan2x graph = 1 mark

graph of tangent line = 1 mark

no penalty if domain extends beyond $\frac{\pi}{4}$

(v)

$$\begin{aligned}
 A &= \int_0^{\pi/8} \tan 2x \, dx - \frac{QR \times PR}{2} \\
 &= \int_0^{\pi/8} \tan 2x \, dx - \frac{\frac{1}{4} \times 1}{2} \\
 &= \left[-\frac{1}{2} \ln(\cos 2x) \right]_0^{\pi/8} - \frac{1}{8} \\
 &= -\frac{1}{2} \left\{ \ln(\cos \frac{\pi}{4}) - \ln(\cos 0) \right\} - \frac{1}{8} \\
 &= -\frac{1}{2} \left\{ \ln\left(\frac{1}{\sqrt{2}}\right) - \ln(1) \right\} - \frac{1}{8} \\
 &= -\frac{1}{2} \left\{ -\frac{1}{2} \ln 2 - 0 \right\} - \frac{1}{8} \\
 &= -\frac{1}{8} + \frac{1}{4} \ln 2 \\
 \therefore \text{Area} &= \left(-\frac{1}{8} + \frac{1}{4} \ln 2 \right) u^2
 \end{aligned}$$

3

area of triangle = 1 mark

 $\left[-\frac{1}{2} \ln(\cos 2x) \right]_0^{\pi/8} = 1$ mark

correct values for “a” and “b” = 1 mark

This is the end of the Examination Paper